

H2BENCH · PROBLEM STATEMENT

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work with sources)
0.1	2026-09-25	Amish Chadha	Initial scaffold

H2Bench problem statement

Students hear about the hydrogen economy but rarely measure it. Teaching kits are either toys that run a fan for a few minutes or lab systems priced for universities, and neither lets a class follow one quantity of energy from the wall socket, through hydrogen, and back to electricity while measuring every loss on the way.

■ The problem

Governments and industry are committing large sums to hydrogen. The European Union aims to produce 10 Mt of renewable hydrogen a year by 2030 and import another 10 Mt ([European Commission](#)); India targets 5 Mt a year under its National Green Hydrogen Mission ([PIB, Government of India](#)). Yet global demand was almost 100 Mt in 2024 and low-emissions hydrogen was still less than 1 % of production ([IEA, Global Hydrogen Review 2025](#)). Engineers, technicians and policy makers will be asked to judge where hydrogen makes sense, and the core fact they need is quantitative: turning electricity into hydrogen and back loses most of the energy. A 2025 review puts the round trip with fuel cells at 27.4 to 48 % ([Oxford Institute for Energy Studies, 2025](#)).

Schools and colleges struggle to teach this by measurement:

1. **Toy kits hide the numbers.** Sub-watt reversible cells (from about \$60 to \$70, [Horizon Educational](#)) show that the reaction works, but at milliwatt scale the losses in meters, leaks and wiring swamp the result, and nothing is measured by default.
2. **Real components are costly.** A branded 12 W fuel cell stack lists at \$482 to \$576 ([Fuel Cell Store, Fuel Cell Shop](#)) and a 300 mL/min PEM electrolyzer stack near \$1,400 ([MSE Supplies](#)), before any storage, safety or instrumentation.
3. **Safety is the gatekeeper.** Hydrogen burns in air from about 4 to 74 % by volume and ignites with about 0.02 mJ ([US DOE](#)). Many schools will not host a hydrogen rig without detection, ventilation and a limit on how much gas is present.

H2Bench is an open teaching bench at a scale large enough to measure (tens of watts) and small enough to be safe in a classroom (under 10 L of hydrogen, low pressure, interlocked by the portfolio's H2Guard detector).

■ Users and context

User	Need	Context
Secondary and vocational students (about 15 to 19 years)	See and measure each energy conversion; calculate efficiencies from their own data	60 to 90 min practical sessions, groups of 3 to 5, supervised
University engineering students	Polarization curves, Faraday efficiency, gas law measurements, uncertainty analysis	Teaching lab, longer sessions, spreadsheet or notebook analysis
Technical and vocational trainers	A rig that trains safe habits: purge, leak check, interlock test, lockout	Hydrogen technician courses in countries building hydrogen industries
Teachers and lab technicians	Set up in under 15 min, run without specialist support, store safely between lessons	Existing lab benches, shared rooms, limited budgets
School safety officers	Evidence that the gas inventory, pressure and ventilation are bounded	Risk assessment before first use

Operating environment

- **Room:** a school or college lab of about 30 to 150 m³ with mechanical or window ventilation, mains power and an existing bench about 750 mm high.
- **Ambient:** 15 to 30 °C, indoor humidity; the fuel cell class chosen is specified for 5 to 30 °C ambient ([Fuel Cell Store](#)).
- **Water:** deionized or distilled water is needed; PEM stacks are damaged by other water ([MSE Supplies](#)).
- **Use pattern:** one fill and one discharge per lesson, a few lessons a week, tank vented down to about 20 kPa gauge (never to atmospheric, so no air enters) at the end of each day.

■ Constraints

- Garage-buildable prototype, about \$450 USD in `project.yaml`. The first-pass BOM is well above this (see HBN-PRC-001 and `docs/REVIEW.md`).
- Hydrogen inventory under 10 L at 20 °C and 101.3 kPa, stored at 300 kPa gauge or less. No compression beyond what the electrolyzer does itself.
- Electrolyzer under 100 W and fuel cell under 50 W, as the pitch states.
- Operates only when the H2Guard detector, extraction fan and interlock are active.
- Off-the-shelf gas components; no welding or pressure-vessel fabrication.
- Fits on a standard lab bench; carried by two people.
- Open design files (CERN-OHL-S-2.0) and open worksheets so teachers can adapt the lessons.

■ Out of scope

- Hydrogen for heating, cooking or vehicles, and any storage above 1 MPa.
- Alkaline electrolysis with caustic electrolyte (excluded for student safety).
- Metal hydride storage in the first version (kept as an option, see HBN-PRC-001).
- Designing the hydrogen detector and interlock itself (that is H2Guard).
- Certification as laboratory equipment; this is a research and teaching prototype.

■ Prior work

- **Educational fuel cell kits.** Horizon Educational and similar suppliers sell kits from sub-watt reversible cells to 12 to 30 W stacks ([Horizon Educational](#)). They prove the components work in schools; the larger stacks set the cost floor.
- **Laboratory PEM electrolyzers.** Gas chromatography hydrogen generators and stacks from suppliers such as MSE deliver 300 to 500 mL/min at up to about 0.4 MPa, showing that a small stack can fill a low-pressure tank directly ([MSE Supplies](#)).
- **Hydrogen education programs.** The US DOE Hydrogen Program names teachers and students among its target audiences and notes "a general lack of awareness of hydrogen as an energy alternative" ([US DOE Hydrogen Program](#)). EPRI's H2EDGE program develops hydrogen training for the workforce ([EPRI](#)).
- **Round-trip analysis.** Published studies of power-to-hydrogen-to-power give the efficiency range that students should be able to reproduce at small scale ([Oxford Institute for Energy Studies, 2025](#)).

- **H2Guard (this portfolio).** The hydrogen leak detector and ventilation interlock that H2Bench depends on. H2Bench adds no detection of its own.

■ Open questions

- Which curricula to align with first (for example a national vocational hydrogen technician standard or an A-level or IB practical)?
- Will school safety officers accept a 300 kPa gauge buffer tank, or is a near-atmospheric gasbag with a small pump easier to approve?
- Is a class demonstration (teacher-run) or a group practical (student-run) the first use case?